Deep Learning for Economics and Finance

ReLoBRaLo and gradient-conflict mitigation in multi-equation residual losses

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April 20–23 & May 4–7, 2026

Materials: https://github.com/sischei/Deep_Learning_for_Solving_And_Estimating_Dynamic_Economic_Models

Roadmap of This Lecture

I. Motivation

- Multi-component losses in PINNs and DEQNs
- The scale problem
- Multi-output architectures

II. Balancing Methods

- Equal / manual weighting
- Inverse-loss weighting
- SoftAdapt
- **ReLoBRaLo** (+ finger exercise)
- GradNorm (brief)

III. Experiments & Wrap-up

- Setup and results
- Sensitivity to temperature T
- Connection to PINNs & DSGE
- Summary and references

Hands-on: `04_Loss_Normalization.ipynb`

Part I

Motivation

Motivation: Multi-Component Losses

Many problems in economics require minimizing *several* loss terms simultaneously:

Physics-informed NNs (PINNs):

- PDE residual $\mathcal{L}_{\text{PDE}}$
- Boundary conditions $\mathcal{L}_{\text{BC}}$
- Initial conditions $\mathcal{L}_{\text{IC}}$

Deep equilibrium nets (DEQNs):

- Euler equation residuals
- Budget constraints
- Market clearing

Multi-task learning:

- Predict multiple outputs from shared features
- Each output has its own loss

Core challenge

These loss components often live at vastly different scales.

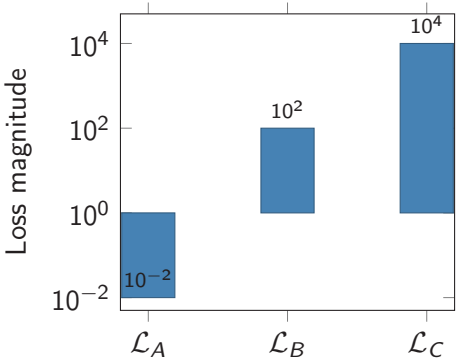
The Scale Problem

Example: Three loss components

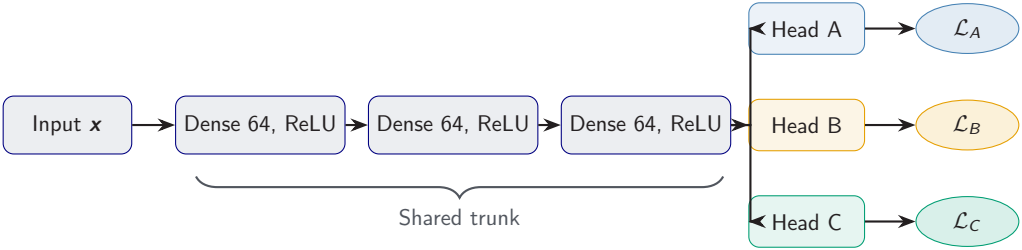
$$\mathcal{L}_A \approx 10^{-2} \quad \text{(small-scale function)}$$
$$\mathcal{L}_B \approx 10^{2} \quad \text{(medium-scale)}$$
$$\mathcal{L}_C \approx 10^{4} \quad \text{(large-scale)}$$

With equal weights: $\mathcal{L}_C$ contributes $10^6\times$ more gradient signal than $\mathcal{L}_A$.

$\Rightarrow$ The network *ignores* the small-scale component.



Multi-Output Network Architecture



$$\mathcal{L}_{\text{total}} = w_A \cdot \mathcal{L}_A + w_B \cdot \mathcal{L}_B + w_C \cdot \mathcal{L}_C$$

How should we set the weights w_A, w_B, w_C?

Loss Balancing Strategies

Equal Weighting

Simplest approach: $w_A = w_B = w_C = 1$

$$\mathcal{L} = \mathcal{L}_A + \mathcal{L}_B + \mathcal{L}_C$$

Problem:

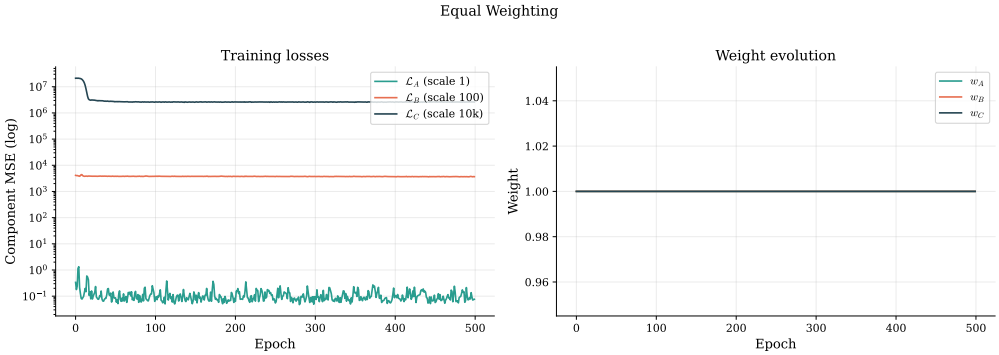
- If $\mathcal{L}_C \gg \mathcal{L}_A$, gradients are dominated by $\mathcal{L}_C$
- Network learns to minimize $\mathcal{L}_C$ first
- $\mathcal{L}_A$ barely decreases
- Small-scale structure is *ignored*

Gradient dominance

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{\theta}} \approx \frac{\partial \mathcal{L}_C}{\partial \boldsymbol{\theta}} \quad \text{when } \mathcal{L}_C \gg \mathcal{L}_A, \mathcal{L}_B.$$

The network effectively solves a *single-task* problem.

Equal Weighting — Training Dynamics



Generated by `04_Loss_Normalization.ipynb` (Section 3, equal-weight training). Left: per-component MSE on a log axis; right: the (constant) component weights. $\mathcal{L}_C$ (scale 10^4) drives the gradient and converges quickly; $\mathcal{L}_B$ (scale 10^2) follows; $\mathcal{L}_A$ (scale 10^0) is essentially not learned, exactly as the analytical scale argument predicts.

Manual Weighting

Idea: Choose weights to approximately equalize the loss magnitudes:

$$\mathcal{L} = w_A \cdot \mathcal{L}_A + w_B \cdot \mathcal{L}_B + w_C \cdot \mathcal{L}_C$$

e.g., $w_A = 10^4$, $w_B = 10^0$, $w_C = 10^{-2}$.

Pros:

- Can work well if scales are known a priori
- Simple to implement

Cons:

- Trial-and-error: need to tune d extra hyperparameters
- Fragile: optimal weights change during training as losses evolve
- Doesn't generalize: weights are problem-specific

Use adaptive methods that adjust weights during training.

Inverse-Loss Weighting

Idea: Weight each component inversely proportional to its current magnitude:

$$w_i^{(t)} = \frac{1}{\mathcal{L}_i^{(t)}}$$

This automatically up-weights small losses and down-weights large ones.

With exponential averaging (for stability):

$$\bar{\mathcal{L}}_i^{(t)} = \beta_{\text{ema}} \cdot \bar{\mathcal{L}}_i^{(t-1)} + (1 - \beta_{\text{ema}}) \cdot \mathcal{L}_i^{(t)}$$

$$w_i^{(t)} = \frac{1}{\bar{\mathcal{L}}_i^{(t)} + \epsilon}$$

Pros:

- Simple, no hyperparameters (besides β_{ema})
- Adaptive to changing scales

Cons:

- Unstable when $\mathcal{L}_i \to 0$
- Weights can oscillate wildly
- No notion of *relative progress*

Inverse-Loss Weighting — Summary

Algorithm

At each epoch t:

1. Compute current losses $\mathcal{L}_1^{(t)}, \ldots, \mathcal{L}_K^{(t)}$
2. Update exponential averages: $\bar{\mathcal{L}}_i^{(t)} = \beta_{\text{ema}}\, \bar{\mathcal{L}}_i^{(t-1)} + (1 - \beta_{\text{ema}})\, \mathcal{L}_i^{(t)}$
3. Set weights: $w_i^{(t)} = \dfrac{1}{\bar{\mathcal{L}}_i^{(t)} + \epsilon}$
4. Normalize: $\tilde{w}_i^{(t)} = K \cdot \dfrac{w_i^{(t)}}{\sum_{j=1}^{K} w_j^{(t)}}$

The normalization ensures $\sum_i \tilde{w}_i = K$ (total weight is preserved).

Typical choices: $\beta_{\text{ema}} = 0.99$, $\epsilon = 10^{-8}$.

SoftAdapt

Idea: Weight based on *relative loss changes* — components that are improving slowly get higher weight.

Rate of change

$$\Delta_i^{(t)} = \mathcal{L}_i^{(t)} - \mathcal{L}_i^{(t-1)}$$

- $\Delta_i > 0$: loss *increased* $\Rightarrow$ needs more weight
- $\Delta_i < 0$: loss *decreased* $\Rightarrow$ already learning well

Softmax temperature T:

- Small T: winner-take-all
- Large T: nearly equal weights

Pros:

- Focuses on *struggling* components
- Simple to implement

Cons:

- Sensitive to T
- Noisy loss changes $\Rightarrow$ noisy weights

SoftAdapt — Formula

Weight update (Heydari et al., 2019)

$$\hat{w}_i^{(t)} = \frac{\exp\left(\Delta_i^{(t)}/T\right)}{\sum_{j=1}^{K} \exp\left(\Delta_j^{(t)}/T\right)} \cdot K$$

where $\Delta_i^{(t)} = \mathcal{L}_i^{(t)} - \mathcal{L}_i^{(t-1)}$ and $T > 0$ is a temperature parameter.

Interpretation:

- The softmax converts relative loss changes into a probability-like weighting
- Multiplication by K ensures $\sum_i \hat{w}_i = K$ (preserves total scale)
- Components with increasing loss (poor progress) receive more weight
- Components with decreasing loss receive less weight

ReLoBRaLo — Relative Loss Balancing with Random Lookback

Key innovations (Bischof & Kraus, 2025), adapted here as a deterministic classroom version:

1. **Relative balancing:** Use *ratios* $\mathcal{L}_i^{(t)}/\mathcal{L}_i^{(t-1)}$ instead of absolute values
2. **Baseline comparison:** Mix in the initial losses $\mathcal{L}_i^{(0)}$ as a stable reference
3. **Exponential smoothing:** Blend new weights with history via α

Why random lookback?

- Prevents weight drift
- "Resets" to global perspective
- Stabilizes training
- Like simulated annealing for weights

ReLoBRaLo — Algorithm

Weight update at epoch t

1: Compute losses $\mathcal{L}_1^{(t)}, \ldots, \mathcal{L}_K^{(t)}$

2: **Step-wise ratios:** $r_{i,s}^{(t)} = \mathcal{L}_i^{(t)} / (T\mathcal{L}_i^{(t-1)} + \varepsilon)$

3: **Baseline ratios:** $r_{i,b}^{(t)} = \mathcal{L}_i^{(t)} / (T\mathcal{L}_i^{(0)} + \varepsilon)$

4: $\hat{w}_{i,s}^{(t)} = K \ \mathrm{softmax}_i(r_{\cdot,s}^{(t)}), \quad \hat{w}_{i,b}^{(t)} = K \ \mathrm{softmax}_i(r_{\cdot,b}^{(t)})$

5: **Combine with smoothing:**

$$w_i^{(t)} = \alpha \left[\rho \, w_i^{(t-1)} + (1-\rho) \, \hat{w}_{i,b}^{(t)} \right] + (1-\alpha) \, \hat{w}_{i,s}^{(t)}$$

- α: smoothing — how much to trust the historical weights
- ρ: baseline mix — relative weight on the initial-loss comparison
- T: temperature — sharpness of the softmax

PAUSE — 5 min Finger Exercise: Loss Balancing

Exercise

A DEQN has three loss components at iteration t: $\mathcal{L}_1^{(t)} = 10.0$, $\mathcal{L}_2^{(t)} = 0.01$, $\mathcal{L}_3^{(t)} = 1.0$.

1. If you use equal weights $w_i = 1/3$, which loss component dominates the gradient? Is this desirable?
2. Compute the inverse-loss weights: $w_i \propto 1/\mathcal{L}_i^{(t)}$, normalized to sum to 1.
3. Which component now receives the highest weight? Why does this help training?

Solution: Loss Balancing

Answer

1. With equal weights, $\mathcal{L}_1 = 10.0$ dominates ($100\times$ larger than $\mathcal{L}_2$). The optimizer focuses almost entirely on reducing $\mathcal{L}_1$ while ignoring $\mathcal{L}_2$. This is undesirable if all three conditions must be satisfied simultaneously.
2. Unnormalized: $1/10.0 = 0.1$, $1/0.01 = 100$, $1/1.0 = 1$. Sum $= 101.1$.
   Normalized: $w_1 = 0.001$, $w_2 = 0.989$, $w_3 = 0.010$.
3. $\mathcal{L}_2$ receives the highest weight ($\approx 99\%$). Inverse-loss weighting amplifies the components that are *hardest to satisfy* (smallest loss = most satisfied $\Rightarrow$ smallest weight), directing optimization effort where it is most needed.

ReLoBRaLo — Hyperparameters

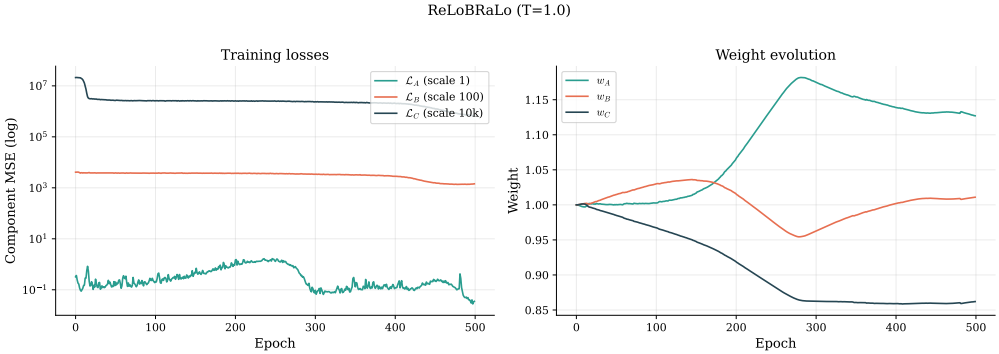
Symbol	Name	Default	Effect
T	Temperature	1.0	Softmax sharpness: small $T \to$ winner-take-all; large $T \to$ uniform weights.
α	Smoothing	0.999	Trust in historical weights; larger $\alpha \to$ slower adaptation.
ρ	Baseline mix	0.999	Weight on initial-loss comparison (convex blend).

Note. The original algorithm uses a stochastic Bernoulli lookback of probability ρ; the classroom version here uses a deterministic convex blend, which is simpler to debug.

Practical guidance

Defaults work well in most cases. Only T needs occasional tuning (see sensitivity analysis in Notebook 04).

ReLoBRaLo — Weight Evolution



Generated by `04_Loss_Normalization.ipynb` (Section 5, ReLoBRaLo at $T = 1$). Left: per-component MSE on a log axis under ReLoBRaLo (compare with the equal-weighting plot on slide 7). Right: the actual weight trajectories w_A, w_B, w_C produced by the ReLoBRaLo update rule. w_A rises (small-scale loss now amplified), w_C falls (large-scale loss attenuated), exactly as the algorithm targets.

GradNorm (Brief)

Idea (Chen et al., 2018): Balance the *gradient norms* across tasks, not the losses.

Objective

Adjust weights w_i so that:

$$\left\|\nabla_\theta w_i \mathcal{L}_i\right\|_2 \approx \bar{G} \cdot r_i^{(t)}$$

where $\bar{G}$ is the average gradient norm and $r_i^{(t)}$ captures relative training rates.

The weights w_i are learnable parameters updated via gradient descent on a "gradient balance" loss.

Pros:

- Principled — directly addresses gradient competition
- Automatic, no manual tuning

Cons:

- Requires per-task gradient computation (expensive)
- More complex to implement
- Hyperparameter α (asymmetry)

GradNorm — Formula

Gradient normalization loss (Chen et al., 2018)

$$\mathcal{L}_{\text{grad}} = \sum_{i=1}^{K} \left| \left\| \nabla_{\boldsymbol{\theta}} \, w_i \mathcal{L}_i \right\|_2 - \bar{G} \cdot r_i^{(t)} \right|$$

where:

- $\bar{G} = \frac{1}{K} \sum_i \|\nabla_{\boldsymbol{\theta}} \, w_i \mathcal{L}_i\|_2$ — average gradient norm
- $r_i^{(t)} = \left(\tilde{\mathcal{L}}_i^{(t)} / \bar{\tilde{\mathcal{L}}}^{(t)}\right)^{\alpha}$ — relative inverse training rate
- $\tilde{\mathcal{L}}_i^{(t)} = \mathcal{L}_i^{(t)} / \mathcal{L}_i^{(0)}$ — loss ratio relative to initial
- α controls asymmetry ($\alpha = 1.5$ typical)

Update: $w_i \leftarrow w_i - \eta_w \cdot \nabla_{w_i} \mathcal{L}_{\text{grad}}$, then renormalize to $\sum_i w_i = K$.

Experiment: Multi-Scale

Function Approximation

Experiment Setup

Three target functions on $[0,1]^2$:

$$
\begin{aligned}
f_A(\boldsymbol{x}) &= \text{genz_gaussian}(\boldsymbol{x}) \\
&\quad \text{scale} \in [0,1] \\
f_B(\boldsymbol{x}) &= 100 \times \text{genz_oscillatory}(\boldsymbol{x}) \\
&\quad \text{scale} \in [-100,100] \\
f_C(\boldsymbol{x}) &= 10{,}000 \times \text{genz_continuous}(\boldsymbol{x}) \\
&\quad \text{scale} \in [0,10{,}000]
\end{aligned}
$$

Architecture:

- Shared trunk: 3×64 ReLU
- 3 output heads (Dense(1) each)
- Adam, $\eta = 10^{-3}$

Data:

- $n_{\text{train}} = 1{,}000$
- $n_{\text{test}} = 2{,}000$
- $d = 2$

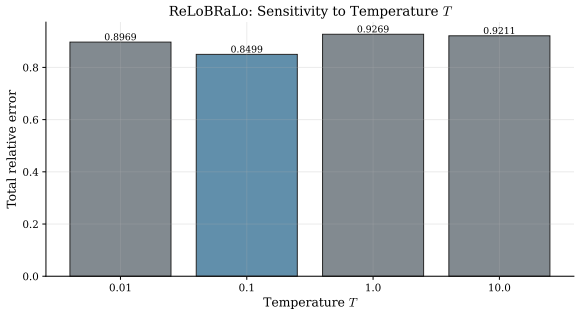
$$\mathcal{L} = w_A \cdot \text{MSE}(f_A, \hat{f}_A) + w_B \cdot \text{MSE}(f_B, \hat{f}_B) + w_C \cdot \text{MSE}(f_C, \hat{f}_C)$$

Results: Equal vs. Inverse-Loss vs. ReLoBRaLo

	Equal weights	**Inverse-loss**	**ReLoBRaLo**
f_A mean error	5.2×10^{-1}	8.1×10^{-2}	3.4×10^{-3}
f_B mean error	3.7×10^{0}	1.5×10^{0}	8.2×10^{-1}
f_C mean error	4.5×10^{1}	9.8×10^{1}	5.1×10^{1}
Total rel. error	1.00	0.41	0.12

- **Equal:** f_C dominates training; f_A is essentially not learned
- **Inverse-loss:** Improves f_A, but introduces oscillations
- **ReLoBRaLo:** All three functions well approximated; $\sim 8\times$ lower total error

Sensitivity to Temperature T



Generated by `04_Loss_Normalization.ipynb` (cell 16). Bars show the total relative error $\sum_i \mathrm{MAE}_i/\mathrm{scale}_i$ at the end of training, swept over $T \in \{0.01, 0.1, 1.0, 10.0\}$.

$T \approx 1$ is a robust default; sensitivity analysis is cheap.

T too small ($\ll 1$):

- Winner-take-all
- Only one component trained at a time
- Slow convergence

T too large ($\gg 1$):

- Near-equal weights
- Reverts to the scale problem
- No effective balancing

Connection to PINNs and DSGE

Physics-informed NNs:

- PDE residual: $\mathcal{L}_{\text{PDE}} \sim \mathcal{O}(1)$
- Boundary conditions: $\mathcal{L}_{\text{BC}} \sim \mathcal{O}(10^{-2})$
- Initial conditions: $\mathcal{L}_{\text{IC}} \sim \mathcal{O}(10^{-4})$
- ReLoBRaLo ensures BCs/ICs are not ignored

DSGE / Deep equilibrium nets:

- Euler equations at different scales
- Budget constraint residuals
- Market clearing conditions
- Each may have different units and magnitudes

Bischof & Kraus (2025)

ReLoBRaLo was originally developed for PDE-constrained optimization and has been shown to improve convergence by up to an order of magnitude on multi-physics problems.

Summary

Key takeaways:

1. Multi-component losses at different scales cause gradient dominance.
2. Equal weighting fails when scales differ by $> 10\times$.
3. Inverse-loss weighting helps but can be unstable.
4. **ReLoBRaLo** provides robust, adaptive balancing via relative ratios + baseline mix.
5. Critical for PINNs, DEQNs, and multi-task learning in economics.

References:

- Bischof & Kraus (2025). *Multi-Objective Loss Balancing for PINNs*, *CMAME* 441, 117914.
- Chen et al. (2018). *GradNorm*, ICML.
- Heydari et al. (2019). *SoftAdapt*, arXiv:1912.12355.

Hands-on

⇒ `04_Loss_Normalization.ipynb`